Course Syllabus

Course Name: Physics I
Course Code: vp150/PHYS1500J (section 2)

Course Pre-requisites
Calculus II (vv116) or Applied Calculus II (vv156) or Honors Mathematics II (vv186)

Textbook
Hugh D. YOUNG, Roger A. FREEDMAN, *University Physics* (15th edition)

Instructor
Mateusz KRZYZOSIAK (mk.ji@gmx.com)
Office hours: Tue 14.10-15.15, Thu 14.10-15.15; and by appointment
Office: 522

Teaching Assistants

HU Hongzheng (email: huhongzheng@sjtu.edu.cn; recitation class: TBA; office hours: TBA)
MENG Yangrui (email: mari_est@sjtu.edu.cn; recitation class: TBA; office hours: TBA)
TANG Zenghao (email: sjtuzh0923@sjtu.edu.cn; recitation class: TBA; office hours: TBA)

A detailed schedule for recitation sessions and office hours will be announced on Canvas.

Grading Policy

Homework 25% = 13% (Mastering Physics) + 12% (paper homework)
On-line activities (15%)
Midterm Exam (30%)
Final Exam (30%)

Note. In past editions of this course, the median course grade has been around “B”.

Academic Integrity

Lectures

Students are encouraged to read the relevant chapters in the textbook ahead of the lecture. Students are required to read and review the relevant chapters after the lecture. Lecture notes will be available on Canvas. Students are expected to attend lectures.

Recitation Classes

Weekly recitation sessions in smaller groups will be led by teaching assistants. Recitation classes will focus mostly on problem solving and discussion. Students are expected to attend and actively participate in the recitation sessions.

Homework

Two types of homework will be assigned: on-line assignments in Mastering Physics and paper homework in the form of problem sets to be solved by each student individually. Discussion of homework problems with other students is allowed and encouraged at the level of general ideas, not specific solutions.
Mastering Physics homework: Problem sets will have a due date assigned, by which date the homework has to be submitted electronically in the Mastering Physics system. Late homework in Mastering Physics cannot be accepted.
Paper homework: Problem sets will have a due date assigned, by which date the homework has to be submitted (information about the submission procedure will be posted on Canvas). All work must be shown, and must be clearly legible. Please plan your time well, as late paper homework will be accepted with the following late-submission penalties: 20% for submissions after the due time but no later than 24 hrs and 40% for submissions 24-48 hrs after the due time. No credit will be given for homework submitted more than 48 hrs after the due time.

Exams

There will be one midterm exam and one final exam as listed in the class schedule. The form of the exams will be announced at least one week before the exam date. The use of a non-electronic English-Chinese dictionary will be allowed during the exams.

Honor Code

Oral discussion of homework problems with other students is allowed and encouraged at the level of general ideas, not specific solutions. It is not allowed to show any written work to other students. If any references to academic textbooks or research journals are made, they should be properly identified with the bibliographical data. No references to Wikipedia entries are allowed. For on-line teaching, additional regulations listed in the *Addendum to the Honor Code for Online courses* apply.

Course description and detailed teaching schedule

Physics I (vp150) is the first part of a course in general physics. It covers classical mechanics, including elements of fluid mechanics, and gravitation (see the schedule below). The goal of the course is to provide students with a basic understanding of nature, that will allow them to apply the laws of physics to formulate and solve engineering problems. The approach used in this course emphasizes patterns and principles relating various phenomena observed in the nature.

Teaching Schedule

week	date	topic	textbook sections
1	May 13-19	nature of physics; physical quantities; kinematics: motion in one dimension	1, 2
2	May 20-26	kinematics: motion in two and three dimensions Newton's laws of motion and their applications	3, *
3	May 27- Jun 2	Newton's laws of motion and their applications	4, 5
4	Jun 3-9	periodic motion	14*
5	Jun 10-16	dynamics in non-inertial frames of reference; work and kinetic energy	4, 5, *, 6
6	Jun 17-23	potential energy and conservation laws	7
7	Jun 24–30	**midterm exam** momentum, impulse, and collisions	8
8	Jul 1-7	rigid body dynamics and angular momentum	9, 10
9	Jul 8-14	rigid body dynamics and angular momentum	9, 10
10	Jul 15–21	equilibrium and elasticity; elements of fluid mechanics	11, 12
11	Jul 22–28	gravitation	13
12	Jul 29 -Aug 4	mechanical waves and sound	15, 16 (selected topics)
13	Aug 5-11	**final exam**	

* additional materials will be provided

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